

DentalExplain

A Dental Diagnosis Expert System

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Abstract

DentalExplain is a native desktop expert system for assessing common tooth-pain and gum-symptom presentations from predefined consultation answers and dentist-supplied examination findings. The application supports five candidate conditions: dental caries, reversible pulpitis, symptomatic irreversible pulpitis, gingivitis and periodontitis. Java Swing provides the interface, while SWI-Prolog 10.0.2 performs validation, adaptive question routing and explicit forward chaining through JPL.

The knowledge base contains 30 authored domain facts and 25 production rules. A separate catalogue defines 30 questions: four setup fields, nine reported-symptom questions and 17 examination questions. The consultation uses two adaptive questionnaire steps, preserves applicable answers during Back navigation and excludes answers when their parent makes them inapplicable. Inference evaluates the complete condition catalogue and permits coexisting candidates. Unknown information remains distinct from explicit absence.

Twenty synthetic acceptance cases and 43 Prolog checks pass. Java integration and interface checks, together with bundled-runtime verification, pass on Apple Silicon macOS, Windows x64 and Ubuntu 24.04 x64. Manual desktop verification was performed on macOS; Windows and Linux verification was automated. The submission includes runnable platform packages and interface and Prolog source. Kushala Jayawickrama, a final-year fifth-year Dental Surgery undergraduate at the University of Peradeniya, is the project's human expert. The expert questionnaire is drafted for confirmation, and clinical knowledge review remains pending.

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1 Introduction

1.1 Background

An expert system represents a specific area of knowledge as reusable facts and rules and applies an inference procedure to a supplied case. DentalExplain uses this approach for a defined set of dental presentations. Rather than accepting unstructured descriptions, it collects reported symptoms and examination findings through predefined controls, then evaluates explicit production rules.

Dental symptoms can overlap, while a condition such as early caries may be present without reported pain [1]. A useful consultation therefore needs both symptom information and objective findings.

DentalExplain keeps these groups separate and retains basic caries and periodontal observations even when the user reports no pain or gum symptoms.

1.2 Problem statement

The problem addressed is how to provide a consistent, runnable expert-system consultation for five common dental conditions without ambiguous free-text inputs or an unnecessarily long questionnaire. The system must preserve uncertainty, identify missing or conflicting information, and evaluate evidence without requiring the user to select a diagnosis in advance.

DentalExplain addresses this problem through a shared Prolog question catalogue, adaptive routing and forward fixed-point reasoning. Its outputs are supported candidates and assessment statuses. It does not make an autonomous clinical diagnosis. The intended user is a dentist who enters findings obtained and interpreted outside the application.

2 Domain Definition and Scope

2.1 Specific domain

The domain is dental diagnostic decision support for common tooth pain and gum symptoms across age groups, using reported symptoms, dentition and explicitly supplied dental examination findings. The supported targets are dental caries, reversible pulpitis, symptomatic irreversible pulpitis, gingivitis and periodontitis.

Age is selected as 0-5, 6-12, 13-17, 18-64 or 65-120 years. These are consultation context groups, not diagnostic thresholds. Dentition and root maturity are supplied independently; the engine does not infer them from age.

2.2 Included scope

The consultation collects four setup fields, reported pain and gum symptoms, warning signs, caries observations, relevant pulp/apical findings and periodontal measurements. Pulpal branches distinguish supplied primary, immature permanent and mature permanent tooth findings. Periodontal rules use attachment-loss patterns and exclusion of other local causes [2-6].

The system supports multiple candidates where their premises coexist. It can return missing-input requests, input conflicts, invalid-input status, an outside-scope response or no supported conclusion. A read-only workspace exposes the separate question, fact and rule catalogues.

2.3 Excluded scope

Orthodontic planning, oral cancer diagnosis, treatment prescribing, image interpretation, patient-record persistence, result export and knowledge editing are excluded. Facial swelling, drainage or fever leads to the existing outside-scope response rather than a diagnosis within this five-condition catalogue. A full dental differential diagnosis and periodontal staging/grading are not implemented.

3 Knowledge Acquisition

3.1 Human expert and knowledge sources

The project's human expert is Kushala Jayawickrama, a final-year fifth-year Dental Surgery undergraduate at the University of Peradeniya. The expert's role is to clarify domain terminology, discuss clinically relevant questions, review provisional fact/rule combinations and assess expected responses, including primary and immature permanent tooth presentations. This qualification is recorded as supplied; the report does not describe the expert as a qualified dentist.

The implemented knowledge draws on NIDCR caries information, AAE diagnostic terminology, AAPD pediatric pulp guidance, AAP gum-disease information and EFP periodontal classification references [1-6]. These sources provide provenance for the authored knowledge; citing them does not imply their authors approved DentalExplain.

3.2 Questionnaire and rule traceability

The human expert questionnaire is included in Appendix B. It contains 15 questions covering the five supported conditions, relevant examination findings, dentition, root maturity, uncertainty, conflicting evidence, warning signs and expected outputs. It is drafted for confirmation. No interview responses, dates, signatures or completed approval are invented.

Each fact and production rule has an identifier, a source and a pending review status. Appendix C lists the implemented catalogue. For example, caries questions link to r01-r04, pulpal criteria to r05-r17 and periodontal criteria to r18-r25. This makes source interpretation and proposed expert review identifiable without adding an explanation facility to the application.

The acquisition process is source review, authored Prolog records, expert discussion, review of rule combinations and expected cases, revision where needed, and repeat verification. Recorded software test passes establish implementation behavior; clinical validation remains a separate review activity.

4 Expert System Architecture

4.1 Anatomy of the system

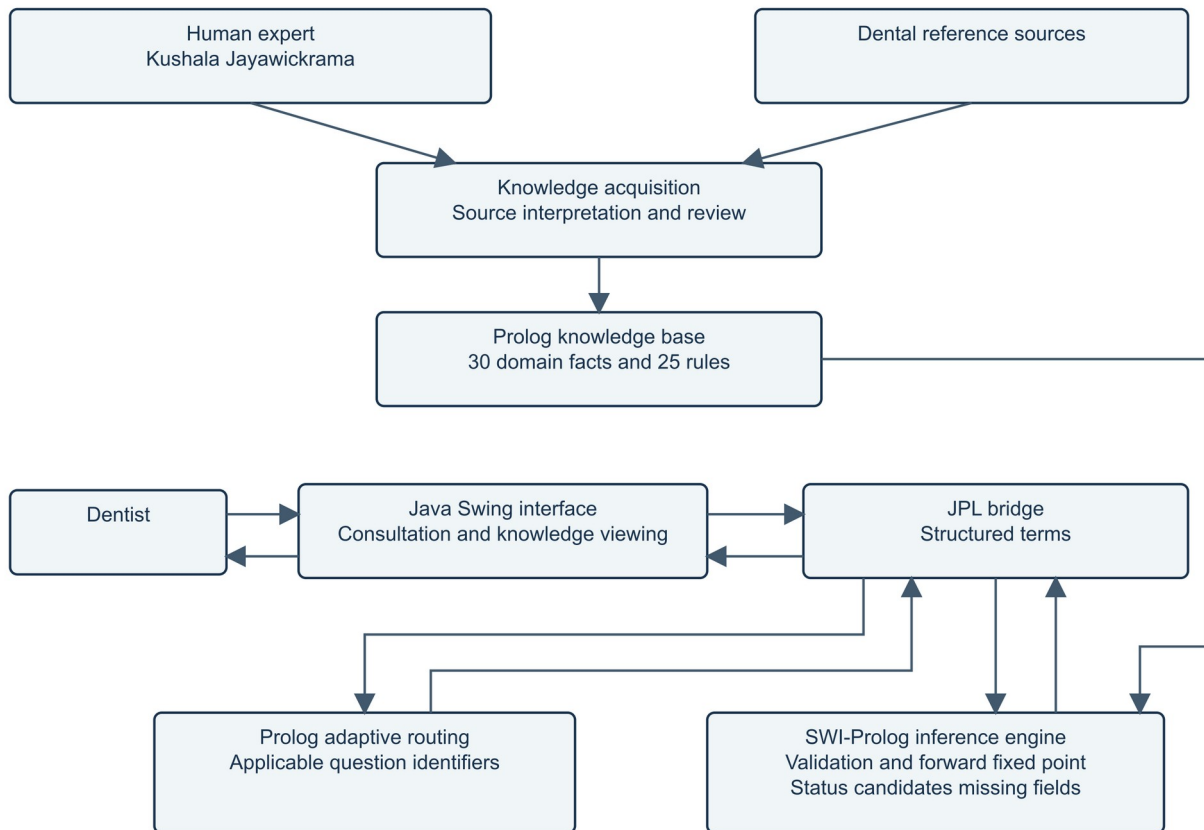


Figure 1 Expert system block diagram

The human expert and documentary sources inform knowledge acquisition. Authored Prolog knowledge is consulted by the embedded inference engine. The dentist supplies controlled selections through Java Swing; JPL exchanges structured terms with Prolog. Routing returns applicable question identifiers, while assessment returns status, candidates, missing fields and messages to the interface.

4.2 Implemented components

Component	Responsibility
Human expert and knowledge acquisition	Clarify domain concepts and review authored knowledge and expected outcomes.
SWI-Prolog knowledge base	Store 30 domain facts, 25 rules and the separate question catalogue with provenance.
Java Swing interface	Present setup, two questionnaire steps, results and the read-only catalogue.
JPL bridge	Exchange atoms, numbers, lists and compound terms with embedded Prolog.
Adaptive routing	Determine applicable questions from setup, earlier answers and measurements.
Forward inference	Validate observations, derive supported patterns/candidates and propagate unresolved prerequisites.
Knowledge workspace	Display searchable Questions, Facts and Rules tabs without editing.

Consultation observations are passed into each inference call. Intermediate deductions remain within inference processing. The architecture includes no separate working-memory component or explanation facility.

5 Knowledge Representation

5.1 Facts

The 30 authored domain facts are reusable statements represented by domain_fact/5. Each record contains an identifier, a structured domain term, a description, a source identifier and a review status. One example is:

```
domain_fact(f07, applicable_dentition(carries,both),  
  'Caries can occur in primary and permanent teeth.',  
  nidcr, pending).
```

Some facts are explicitly queried through kb(...) premises. Others describe clinical concepts and provenance in the knowledge catalogue. They are not 30 patient observations. Question schemas, metadata, test data and per-call deductions do not inflate the authored fact count.

5.2 Production rules

The 25 rule/5 records contain an identifier, premise list, conclusion, source and review status. Equality and numeric comparisons test supplied findings; derived(...) requires an intermediate deduction; kb(...) checks an authored domain term. For example:

```
rule(r01, [eq(cavity,yes),eq(soft_tissue,yes)],  
  carious_lesion, nidcr, pending).  
rule(r04, [derived(carious_lesion),  
  kb(applicable_dentition(carries,both))],  
  candidate(carries), nidcr, pending).
```

Rules r01-r04 concern caries. Rules r05-r17 form provoked-pain and pulpal patterns and apply tooth/root-specific branches. Rules r18-r25 form gingival inflammation and periodontal destruction patterns. The complete premises and conclusions appear in Appendix C.

5.3 Questions and consultation observations

The separate catalogue defines 30 questions with identifiers, stage membership, control types, labels, allowed values and applicability. Dropdowns are non-editable; radio and checkbox answers use question-specific wording. Labels map to stable Prolog atoms or predefined numbers. A submitted answer is represented as obs(Key,Value).

Unknown is the default and is not treated as No. Not applicable is separate but cannot establish a required premise. For checkbox groups, None explicitly records absence of all listed items. Selecting an item establishes only that finding; other unchecked items remain Unknown. Combined gum and warning-sign answers expand to the internal clinical observation identifiers before inference.

6 Inference Method

6.1 Forward chaining and justification

Forward chaining is the only implemented method. A consultation begins with evidence and seeks every supported candidate in the five-condition catalogue. This suits data-driven reasoning: the user need not nominate a suspected diagnosis, and different rule branches can support coexisting candidates. The decision replaces the earlier plan to provide both chaining methods.

The engine begins with an empty deduction set, scans every production rule and adds conclusions whose premises are satisfied. It sorts and deduplicates the combined set and repeats until the set stops changing. The finite conclusion vocabulary and duplicate prevention guarantee termination for this rule base.

6.2 Worked dental example

Consider TC06: an adult group, permanent tooth, mature roots, a supplied carious lesion, cold/sweet-provoked pain that stops promptly, no spontaneous or sleep-interrupting pain, no percussion/apical findings, no warning signs and a brief thermal response.

Forward pass	Newly supported deductions	Relevant rules
1	carious_lesion and provoked_pain	r02, r05 and r06
2	candidate(caries) and brief_provoked_pain	r04 and r08
3	reversible_pattern	r09
4	candidate(reversible_pulpitis)	r12
5	No further conclusions; fixed point reached	No new rule conclusion

The returned candidates are Dental caries and Reversible pulpitis. Explicit negative findings are premises in r09, so an unanswered negative finding cannot silently satisfy that rule. This walkthrough describes the algorithm in the report; the interface does not display fired-rule traces.

6.3 Missing information and assessment statuses

After confirmed deductions settle, a second forward pass propagates unresolved prerequisite sets through unblocked rules until stable. A known contradictory premise blocks a rule and its missing-field requests. Unknown or Not applicable evidence remains unusable. Missing requests for internal checkbox findings map back to their public question, and hidden follow-ups point to an unanswered parent first.

Input validation, contradiction checks and outside-scope checks precede ordinary assessment. Missing age group requests completion. Otherwise the engine returns supported candidates, additional information needed, or no supported condition. Candidate outputs can also include missing findings for other unresolved conditions; this does not retract the supported candidates.

7 System Design and Implementation

7.1 Technology and execution boundary

Java 21 and Swing provide the native desktop interface. SWI-Prolog 10.0.2 performs reasoning through matching JPL Java/JNI libraries. Java Swing was selected for standard controls and a shared interface codebase. JPL still requires platform-specific native dependencies [7]. Windows and macOS application images use Java packaging tools [8]. The expert-system implementation uses Java, Prolog and shell scripts, with no Python inference or interface implementation.

The Java bridge constructs typed terms instead of concatenating user text into Prolog queries. The public assessment and routing interfaces are:

```
assess(Observations,  
      result(Status,CandidateIds,MissingQuestionIds,Messages)).  
active_questions(Observations,Stage,QuestionIds).
```

Java calls Bridge.assess(answers) and Bridge.activeQuestions(answers,stage). Prolog validates identifiers, allowed values, duplicate keys and list shapes at the boundary. Catalog operations expose questions, facts, rules, condition labels, sources and knowledge version.

7.2 Adaptive consultation and lifecycle

Setup collects age group, dentition, tooth type and region. Step 1 collects reported symptoms; pain follow-ups appear when tooth pain is reported. Step 2 includes basic caries observations and periodontal measurements even without symptoms, then reveals follow-ups according to supplied findings. Mature-root thermal fields are shown only for an applicable painful permanent tooth with mature roots. Known warning signs or the jaw-only pattern bypass unnecessary examination questions.

Prolog controls applicability. Java clears and excludes inapplicable child answers, while preserving applicable selections across Back navigation. Initialization, routing and reasoning run on one background executor. Swing updates occur on the event-dispatch thread. Generation tokens discard callbacks from obsolete edits, navigation or reset [9].

Results provide New consultation as the only action. It clears selections and results and restores defaults. Back navigation remains available before assessment. Knowledge viewing is read-only; no editing or result-saving dialog is provided.

7.3 Executable and runtime dependencies

Application version 1.1.0 uses knowledge version 0.3.0. The submission contains a Windows x64 executable image, an Apple Silicon macOS app and a bundled Java package for Ubuntu 24.04 x64 desktops. Each retains Java, SWI-Prolog/JPL, knowledge files and legal notices. Users must keep complete application folders together.

The root launchers open those packages without compilation or separate Java/Prolog installation. The Java application JAR is shared code, but cannot run alone without its accompanying dependencies. Windows signing and macOS notarization remain pending; native platform packages are distinct from an installer.

7.4 Welcome and consultation setup

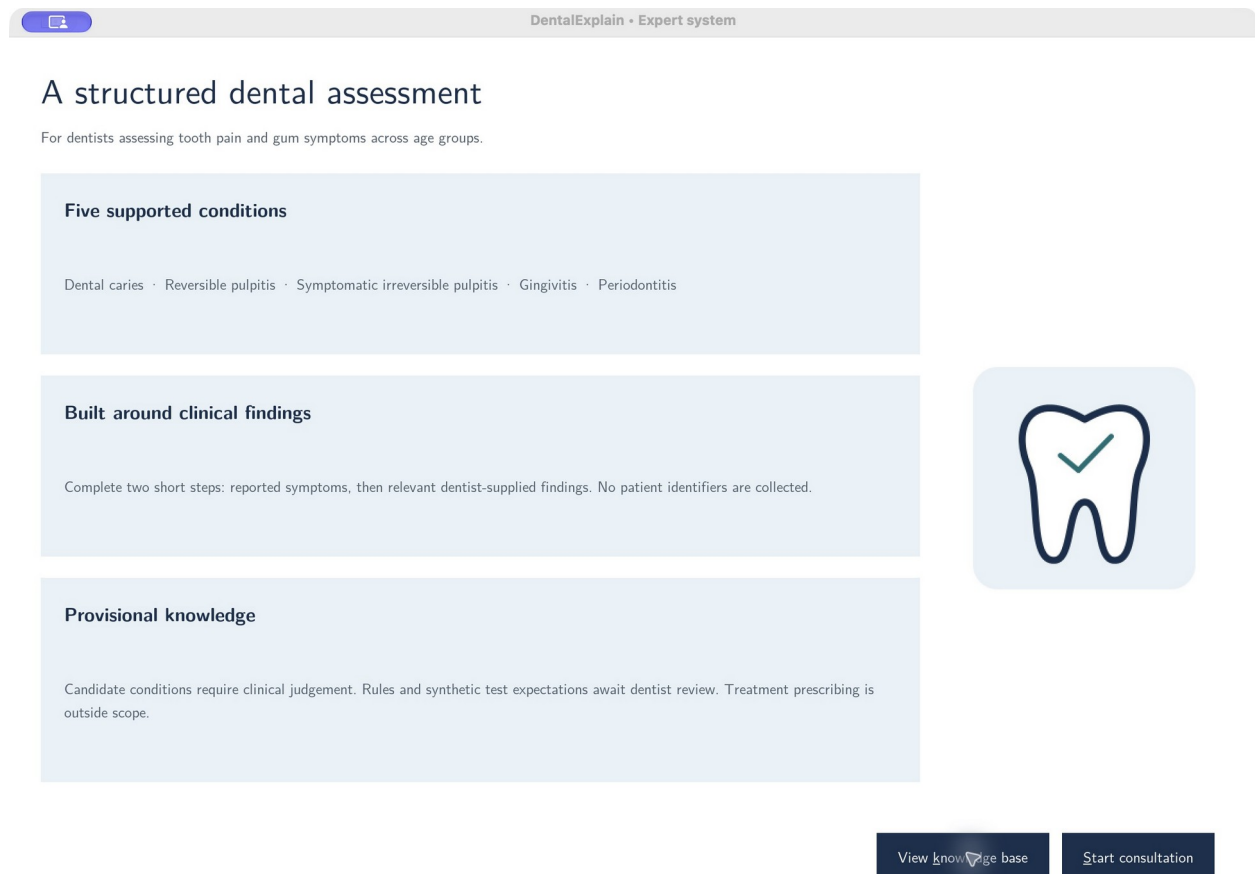


Figure 2 Current welcome screen

The welcome screen offers Start consultation and View knowledge base, identifies the supported conditions and describes the consultation flow.



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01 / Consultation setup

Record the age group and affected dentition for the assessment.

Age group	Dentition
18-64 years	Permanent
Affected tooth type	Affected region
Permanent tooth	Single tooth

[Welcome](#)[Continue to symptoms](#)

Figure 3 Current setup with predefined age and dentition selections

Setup uses four non-editable dropdowns. Age group does not automatically determine dentition or diagnosis. The form retains its white background, internal padding and consistent Latin Modern Sans typography.

7.5 Reported symptoms and relevant findings

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02 / Reported symptoms

Questionnaire • Step 1 of 2 • Select the reported findings. Unchecked items remain unknown.

Reported symptoms

Is tooth pain present?

☐ Unknown

☒ Tooth pain reported

☐ No tooth pain reported

☐ Not applicable

What triggers the tooth pain?

☐ Unknown

☒ Cold

☒ Sweet

☐ Hot

☐ Biting

☐ None

☐ Not applicable

Pain after the trigger stops

Unknown

Spontaneous tooth pain?

☐ Unknown

☐ Pain starts without a trigger

☒ No spontaneous pain

☐ Not applicable

Back to setup

Continue to findings

Figure 4 Pain selection reveals question-specific symptom follow-ups

Step 1 separates reported symptoms from examination findings. Reporting tooth pain reveals trigger, persistence, spontaneous, sleep and biting questions. Checkbox special answers remain mutually exclusive with findings.

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03 / Relevant dental findings

Questionnaire · Step 2 of 2 · Follow-ups adapt to your symptoms and examination measurements.

Dentist-supplied tooth findings

Cavitated lesion observed?

☐ Unknown

☐ No cavity observed

☒ Cavity observed

☐ Not applicable

Softened tooth tissue observed?

☐ Unknown

☐ No softened tissue observed

☒ Softened tissue observed

☐ Not applicable

Decay-related surface discoloration?

☒ Unknown

☐ No decay-related discoloration

☐ Decay-related discoloration observed

☐ Not applicable

Percussion or palpation tenderness?

☒ Unknown

☐ Not tender

☐ Tender to percussion or palpation

☐ Not applicable

Dentist-interpreted coronal carious lesion?

Back to symptoms

Assess presentation

Figure 5 Dentist-supplied findings in the second questionnaire step

Step 2 presents applicable caries, pulp/apical and periodontal findings. Long forms scroll, while Back and Assess actions remain available outside the scroll area. Uncertainty keeps potentially relevant follow-ups available.

7.6 Results and knowledge workspace



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04 / Assessment result

Candidate conditions can coexist. Results do not establish an autonomous diagnosis.

Supported candidate conditions

- Dental caries

Candidate conditions only. Clinical knowledge review is pending.

Missing or unusable findings:

- Sites bleeding on probing
- Maximum buccal/oral attachment loss
- Maximum interdental attachment loss
- Interdental loss at two nonadjacent teeth?
- Could another local cause account for the attachment loss?
- Plaque or calculus present?
- Maximum probing depth
- Previous periodontal destruction?
- Buccal/oral loss at two teeth?

New consultation

Figure 6 Supported caries candidate from a synthetic consultation

Results display supported candidates and any additional requested findings. New consultation is the only result action. The screenshot represents a synthetic symptom-free caries assessment, separate from the pain-follow-up illustration.



Knowledge workspace

Read-only catalogue. Domain facts, production rules and consultation questions have separate counts.

Questions (30) Facts (30) Rules (25)			
Search questions (30)			
ID	Section / type	Question / allowed mappings	Source / i
age_group	setup / select	Age group [option(unknown, 'Unknown'), option(young_child, '0-5 yea...	https://www.aapd.org/media/Policies_Guideli
dentition	setup / select	Dentition [option(unknown, 'Unknown'), option(primary, 'Primary'), o...	https://www.aapd.org/media/Policies_Guideli
tooth_t...	setup / select	Affected tooth type [option(unknown, 'Unknown'), option(primary, 'Pr...	https://www.aapd.org/media/Policies_Guideli
region	setup / select	Affected region [option(unknown, 'Unknown'), option(single_tooth, 'Si...	https://www.aapd.org/media/Policies_Guideli
tooth_p...	symptoms / radio	Is tooth pain present? [option(unknown, 'Unknown'), option(yes, 'Toot...	https://www.aae.org/specialty/wp-content/up
triggers	symptoms / multi	What triggers the tooth pain? [option(unknown, 'Unknown'), option(co...	https://www.aae.org/specialty/wp-content/up
persiste...	symptoms / select	Pain after the trigger stops [option(unknown, 'Unknown'), option(brief...	https://www.aae.org/specialty/wp-content/up
spontan...	symptoms / radio	Spontaneous tooth pain? [option(unknown, 'Unknown'), option(yes, 'P...	https://www.aae.org/specialty/wp-content/up
sleep_pain	symptoms / radio	Tooth pain interrupts sleep? [option(unknown, 'Unknown'), option(yes,...	https://www.aae.org/specialty/wp-content/up
biting_...	symptoms / radio	Pain when biting? [option(unknown, 'Unknown'), option(yes, 'Pain on ...	https://www.aae.org/specialty/wp-content/up
Select a row to read its full content and source.			



Welcome

Figure 7 Read-only knowledge workspace with separate catalogue tabs

Questions, Facts and Rules have separate counts. Search filters the active table, and selecting a row displays its full content and source. The workspace provides inspection rather than knowledge editing.

8 Testing and Evaluation

8.1 Testing strategy

The acceptance catalogue contains 14 diagnostic cases and six edge cases. Inputs are synthetic and mapped to predefined selections. Diagnostic cases require the target candidate without forcing mutually exclusive diagnoses. Edge cases cover missing age, invalid numeric age-group input, incomplete evidence, contradictory pain inputs, outside-scope presentation and reset. Full case inputs, expected responses and actual engine results appear in Appendix D.

Additional Prolog checks cover valid choices, malformed/removed identifiers, checkbox exclusivity, uncertainty, age groups, termination, duplicate prevention, blocked rules, routing and coexisting candidates. Java/JPL checks assess the same cases and 14 routed diagnostic cases. Swing checks cover 30 control schemas, Back navigation, hidden-answer clearing, results, reset and stale callbacks.

8.2 Automated results

Check	macOS ARM64	Windows x64	Ubuntu 24.04 x64
20 forward acceptance cases	PASS	PASS	PASS
43 Prolog checks	PASS	PASS	PASS
20 JPL cases and 14 routed diagnostic cases	PASS	PASS	PASS
Controlled inputs and consultation lifecycle checks	PASS	PASS	PASS under Xvfb
Bundled Java launch from a path with spaces	PASS	PASS	PASS
Native launcher initialization	PASS app	PASS exe	Java launcher used

These results are recorded in the successful cross-platform workflow on 29 September 2026 [10]. The current source retains the tested engine and application behavior. The report-data export separately assesses all 20 fixtures and records the actual candidates, messages and missing identifiers in Appendix D.

8.3 Desktop and submission verification

Manual desktop verification on macOS includes consultation navigation, result display, knowledge viewing and New consultation. Windows and Linux desktop behavior was checked automatically on hosted runners; a separate manual Windows/Linux consultation is not claimed. Screenshots in this report were captured from the current installed macOS application using synthetic selections.

The submission assembly verifies source inclusion, report files, three root launch scripts, complete runtime directories, executable permissions and SHA-256 checksums. Launch checks use extracted folders containing spaces. Recorded software results do not establish diagnostic accuracy in real patients; clinical knowledge and expectations remain pending review.

9 Conclusion

DentalExplain implements a runnable native expert system combining Java Swing with embedded SWI-Prolog through JPL. Its 30-question adaptive consultation collects controlled evidence in two steps, while 25 production rules and 30 authored domain facts support five candidate conditions through forward chaining.

The implementation preserves uncertainty, supports coexisting candidates and returns defined responses for incomplete, conflicting and unsupported inputs. The submission provides source, a platform-specific runnable package for each supported operating system and a user manual. Automated verification passes across the three target platforms, with manual macOS inspection. The human expert is identified and the drafted review questionnaire is supplied in Appendix B; clinical review remains pending.

References

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- [10] DentalExplain. Verified cross-platform workflow run, 29 September 2026, source commit 777ab83. <https://github.com/vishwajayawickrama/dental-expert-system/actions/runs/36535968417>
- References accessed on 29 September 2026. Source identifiers used in Appendix C map as follows: nidcr [1], aae [2], aapd [3], aap [4], efp [5], efp_g [6].

Appendix A User Manual

A.1 Starting the distributed application

Extract DentalExplain-submission.zip completely before opening the app. Choose the launcher for your computer. The bundled runtimes mean Java and SWI-Prolog do not need separate installation. Keep the applications folder and all its contents intact; do not move only the executable or JAR.

Computer	Launch from the extracted ZIP root
Windows x64	Double-click Open-Windows.cmd. Alternatively, open DentalExplain.exe inside the applications/windows/DentalExplain folder.
Apple Silicon macOS	Double-click Open-macOS.command, or open applications/macos/DentalExplain.app. The app may be copied intact to Applications.
Ubuntu 24.04 x64 desktop	Open a terminal in the extracted folder and run ./Open-Linux.sh. A graphical desktop is required.

If a Unix extraction tool removes executable permissions, run `chmod +x Open-macOS.command` on macOS. On Linux run `chmod +x Open-Linux.sh applications/linux/DentalExplain/runtime/java/bin/java`. Windows packages target x64; Intel Macs and other ARM targets are not provided. Windows signing and macOS notarization are pending; follow the computer owner's software policy if the operating system blocks a downloaded application.

Wait for the welcome screen before starting. An initialization error means the application runtime is unavailable; it is not a clinical result. The root scripts report missing application files. Restore the complete extracted package rather than replacing individual libraries.

A.2 General consultation procedure

Choose Start consultation, supply setup details, complete Reported symptoms, then Relevant dental findings, and choose Assess presentation. Back to setup and Back to symptoms allow changes before assessment. Applicable selections remain; changing a parent clears dependent answers that become inapplicable.

Use Unknown for unavailable information. A negative answer explicitly records absence, while Not applicable cannot establish a diagnostic premise. In checkbox groups, None reported excludes every listed finding. An unchecked item remains Unknown unless None is selected. Use Tab and Shift-Tab to move between controls, arrows for predefined choices and Space for radio/checkbox controls.

A.3 Example consultation for symptom-free caries

This example is synthetic. Choose age group 18-64 years, Permanent dentition, Permanent tooth and Single tooth. Continue to symptoms. Select No tooth pain reported, None reported for gum symptoms and warning signs, and No jaw clicking reported. Pain follow-ups stay hidden.

Continue to findings and select Cavity observed and Softened tissue observed. Record other findings only if actually supplied; leave unavailable information Unknown. Assess presentation. These findings support Dental caries under the implemented rules, and unresolved other conditions may produce additional missing-field requests.


DentalExplain • Expert system

01 / Consultation setup

Record the age group and affected dentition for the assessment.

Age group
18-64 years

Dentition
Permanent

Affected tooth type
Permanent tooth

Affected region
Single tooth

Welcome
Continue to symptoms

Figure A1 Setup for the synthetic consultation

A.4 Example consultation with pain follow-ups

For a painful permanent tooth, report Tooth pain reported in Step 1. Trigger, persistence, spontaneous, sleep and biting questions appear. Select only the supplied findings. In Step 2, provide root maturity; a recorded thermal response becomes available for mature roots. The TC06 values in Appendix D support coexisting caries and reversible pulpitis candidates.

Measurements use predefined dropdowns: depth and attachment loss range from 0 to 15 mm in 0.5 mm steps, and bleeding on probing from 0% to 100% in 5% steps. Do not invent or round a finding just to select a rule-supporting value. Use Unknown for unavailable or unrepresentable measurements.

A.5 Interpreting results and starting again

Outcome	User action
Supported candidate conditions	Interpret the candidates with clinical judgement; more than one can be supported.
Additional information needed	Review the requested findings and start a fresh consultation with available evidence.
Conflicting or invalid selections	Start again with consistent predefined values.
Outside supported scope	Assess the presentation beyond the five-condition catalogue.
No supported conclusion	No rule-supported candidate follows; this does not exclude other dental conditions.

The results screen offers New consultation only. It clears selections and the displayed result. Results cannot be edited or saved from this screen. Closing the app does not store the consultation.

04 / Assessment result

Candidate conditions can coexist. Results do not establish an autonomous diagnosis.

Supported candidate conditions

- Dental caries

Candidate conditions only. Clinical knowledge review is pending.

Missing or unusable findings:

- Sites bleeding on probing
- Maximum buccal/oral attachment loss
- Maximum interdental attachment loss
- Interdental loss at two nonadjacent teeth?
- Could another local cause account for the attachment loss?
- Plaque or calculus present?
- Maximum probing depth
- Previous periodontal destruction?
- Buccal/oral loss at two teeth?

[New consultation](#)

Figure A2 Supported candidate and the New consultation action

A.6 Knowledge viewing and troubleshooting

Choose View knowledge base from Welcome. Questions (30), Facts (30) and Rules (25) are separate read-only tabs. Search filters the active table. Select a row to inspect its content, source and review status; scroll to view long entries.

Problem	Action
Missing runtime or native library	Restore the complete package, keeping application folders together.
Unsupported architecture	Use the supplied macOS ARM64, Windows x64 or Ubuntu x64 package.
Launcher cannot execute on Unix	Restore the executable permissions described in A.1.
Missing age group	Choose one of the five predefined groups before assessment.
Missing findings	Use only available evidence; do not turn Unknown into a negative answer.
Unexpectedly empty knowledge table	Clear the search filter and select the relevant tab.

Appendix B Human Expert Questionnaire

Human expert: Kushala Jayawickrama, final-year fifth-year Dental Surgery undergraduate, University of Peradeniya. Status: drafted for confirmation. The following questions are proposed for the expert discussion; no responses or completed approval are recorded.

B.1 Proposed questions

1. Which reported symptoms and dentist-supplied findings are most useful when assessing common tooth pain and gum symptoms within these five targets? Relevant knowledge: all rule groups.
2. Which combinations of cavitation, softened tissue, discoloration and radiographic findings support a caries candidate, including a painless presentation? Relevant rules: r01-r04.
3. How should brief cold- or sweet-provoked pain be distinguished from pain that persists after the stimulus stops? Relevant rules: r05-r08 and r14.
4. Which explicitly absent findings should be required before supporting the implemented reversible-pulpitis pattern? Relevant rules: r09-r12.
5. How should spontaneous pain, sleep interruption and lingering thermal pain contribute to a symptomatic irreversible-pulpitis candidate? Relevant rules: r13-r17.
6. What changes in question applicability and interpretation are needed for primary teeth compared with permanent teeth? Relevant facts: f07 and f14-f16; rules r10 and r15.
7. How should immature permanent roots affect pulp testing and the required supporting evidence? Relevant facts: f15; rules r11 and r17.
8. For mature permanent teeth, how should a brief, lingering or absent thermal response be interpreted alongside other findings? Relevant rules: r12 and r16.
9. Which combinations of plaque, bleeding, red/swollen gingiva and bleeding-on-probing measurements support gingival inflammation? Relevant rules: r18-r19 and r25.
10. What attachment-loss, probing-depth and previous-destruction findings are required for the implemented intact-periodontium gingivitis candidate? Relevant rule: r20.
11. How should interdental attachment loss at nonadjacent teeth support a periodontitis case definition? Relevant rules: r21 and r23.
12. How should buccal/oral attachment loss, pocket depth, affected-tooth count and alternative local causes be combined? Relevant rules: r22-r24.
13. How should missing, Unknown, Not applicable and contradictory findings affect the assessment and requests for further information? Relevant processing: boundary validation and forward missing-prerequisite propagation.
14. Which warning signs or jaw-only presentations should leave the supported scope, and is the current routing appropriate? Relevant processing: outside_scope checks.
15. Do the 20 synthetic cases and their expected coexisting candidates match the intended rule behavior, and which facts, premises or questions require revision before clinical validation? Relevant evidence: Appendices C and D.

Appendix C Knowledge Catalogue

These records are exported directly from the implemented Prolog catalogue. All 30 facts and 25 production rules retain pending clinical review status. Consultation observations, question schemas and test fixtures are not counted as domain facts.

C.1 Authored domain facts

ID	Domain statement	Source
f01	Caries damages mineralized tooth tissue.	nidcr
f02	Early caries may have no pain.	nidcr
f03	A cavity supports a caries assessment.	nidcr
f04	Softened tooth tissue supports a caries assessment.	nidcr
f05	A dentist-interpreted radiographic lesion can support caries.	nidcr
f06	White, brown or dark surface changes may accompany decay.	nidcr
f07	Caries can occur in primary and permanent teeth.	nidcr
f08	Reversible pulpitis affects vital inflamed pulp.	aae
f09	Brief provoked pain can support reversible pulpitis.	aapd
f10	Spontaneous pain conflicts with this reversible-pulpitis pattern.	aapd
f11	Spontaneous pain can support symptomatic irreversible pulpitis.	aae
f12	Lingering thermal pain can support symptomatic irreversible pulpitis.	aae
f13	Irreversible pulpitis terminology refers to inflamed vital pulp.	aae
f14	Thermal and electric pulp tests are unreliable in primary teeth.	aapd
f15	Thermal and electric pulp tests are unreliable in immature permanent teeth.	aapd
f16	Primary-tooth symptoms may not distinguish irreversible pulpitis from necrosis.	aapd
f17	Cold can provoke pulpal discomfort.	aae
f18	Sweet stimuli can provoke discomfort in decay-related presentations.	nidcr
f19	Heat can provoke pulpal discomfort.	aae
f20	Gingivitis concerns gingival inflammation.	aap
f21	Plaque is associated with gingival inflammation.	aap
f22	Red or swollen gums support gingival inflammation.	aap
f23	Gum bleeding supports gingival inflammation.	aap
f24	Gingivitis on an intact periodontium has no attachment loss.	efp_g
f25	Previous periodontal destruction changes assessment of gingival inflammation.	efp_g
f26	Periodontitis involves loss of tooth-supporting tissue.	efp
f27	Interdental attachment loss at two nonadjacent teeth supports the case definition.	efp
f28	The alternative case definition uses buccal/oral attachment loss of at least 3 mm.	efp
f29	The alternative definition also requires pocket depth greater than 3 mm.	efp
f30	Attachment loss from nonperiodontal causes must be excluded.	efp

C.2 Production rules

Premise notation: eq is equality; gt/gte/lte compare numeric findings; derived requires an intermediate deduction; kb refers to a domain fact. Every listed premise must hold. Internal gum, trigger and warning identifiers come from checkbox expansion.

R01 carious_lesion

IF eq(cavity,yes) AND eq(soft_tissue,yes) THEN carious_lesion.

Source: nidcr; review: pending.

R02 carious_lesion

IF eq(radiographic_caries,yes) THEN carious_lesion.

Source: nidcr; review: pending.

R03 carious_lesion

IF eq(discoloration,yes) AND eq(soft_tissue,yes) THEN carious_lesion.

Source: nidcr; review: pending.

R04 candidate(caries)

IF derived(carious_lesion) AND kb(applicable_dentition(caries,both)) THEN candidate(caries).

Source: nidcr; review: pending.

R05 provoked_pain

IF eq(tooth_pain,yes) AND eq(trigger_cold,yes) AND kb(pulpal_trigger(cold)) THEN provoked_pain.

Source: aae; review: pending.

R06 provoked_pain

IF eq(tooth_pain,yes) AND eq(trigger_sweet,yes) AND kb(pulpal_trigger(sweet)) THEN provoked_pain.

Source: nidcr; review: pending.

R07 provoked_pain

IF eq(tooth_pain,yes) AND eq(trigger_hot,yes) AND kb(pulpal_trigger(hot)) THEN provoked_pain.

Source: aae; review: pending.

R08 brief_provoked_pain

IF derived(provoked_pain) AND eq(persistence,brief) THEN brief_provoked_pain.

Source: aapd; review: pending.

R09 reversible_pattern

IF derived(brief_provoked_pain) AND derived(carious_lesion) AND eq(spontaneous,no) AND eq(sleep_pain,no) AND eq(percussion,no) AND eq(apical_pathology,no) AND eq(facial_swelling,no) AND eq(drainage,no) AND eq(fever,no) THEN reversible_pattern.

Source: aapd; review: pending.

R10 candidate(reversible_pulpitis)

IF derived(reversible_pattern) AND eq(tooth_type,primary) AND kb(unreliable_pulp_test(primary)) THEN candidate(reversible_pulpitis).

Source: aapd; review: pending.

R11 candidate(reversible_pulpitis)

IF derived(reversible_pattern) AND eq(tooth_type,permanent) AND eq(root_maturity,immature) AND kb(unreliable_pulp_test(immature)) THEN candidate(reversible_pulpitis).

Source: aapd; review: pending.

R12 candidate(reversible_pulpitis)

IF derived(reversible_pattern) AND eq(tooth_type,permanent) AND eq(root_maturity,mature) AND eq(thermal_response,brief) THEN candidate(reversible_pulpitis).

Source: aae; review: pending.

R13 irreversible_pattern

IF eq(tooth_pain,yes) AND eq(spontaneous,yes) AND kb(supports(spontaneous_pain,irreversible_pulpitis)) THEN irreversible_pattern.

Source: aae; review: pending.

R14 irreversible_pattern

IF derived(provoked_pain) AND eq(persistence,lingering) AND kb(supports(lingering_pain,irreversible_pulpitis)) THEN irreversible_pattern.

Source: aae; review: pending.

R15 candidate(irreversible_pulpitis)

IF derived(irreversible_pattern) AND derived(carious_lesion) AND eq(tooth_type,primary) AND eq(sleep_pain,yes) THEN candidate(irreversible_pulpitis).

Source: aapd; review: pending.

R16 candidate(irreversible_pulpitis)

IF derived(irreversible_pattern) AND derived(carious_lesion) AND eq(tooth_type,permanent) AND eq(root_maturity,mature) AND eq(thermal_response,lingering) THEN candidate(irreversible_pulpitis).

Source: aae; review: pending.

R17 candidate(irreversible_pulpitis)

IF derived(irreversible_pattern) AND derived(carious_lesion) AND eq(tooth_type,permanent) AND eq(root_maturity,immature) AND eq(sleep_pain,yes) THEN candidate(irreversible_pulpitis).

Source: aapd; review: pending.

R18 gingival_inflammation

IF eq(gum_bleeding,yes) AND eq(plaque,yes) THEN gingival_inflammation.

Source: aap; review: pending.

R19 gingival_inflammation

IF eq(red_gums,yes) AND eq(plaque,yes) THEN gingival_inflammation.

Source: aap; review: pending.

R20 candidate(gingivitis)

IF derived(gingival_inflammation) AND eq(cal_mm,0) AND eq(prior_destruction,no) AND lte(pocket_mm,3) AND kb(intact_periodontium(gingivitis,0)) THEN candidate(gingivitis).

Source: efp_g; review: pending.

R21 periodontal_destruction

IF gt(cal_mm,0) AND eq(nonadjacent_teeth,yes) AND eq(nonperiodontal_causes,no) AND kb(exclude_nonperiodontal_causes) THEN periodontal_destruction.

Source: efp; review: pending.

R22 periodontal_destruction

IF gte(buccal_cal_mm,3) AND derived(deep_pocket) AND eq(two_teeth,yes) AND eq(nonperiodontal_causes,no) THEN periodontal_destruction.

Source: efp; review: pending.

R23 candidate(periodontitis)

IF derived(periodontal_destruction) AND kb(defining_feature(periodontitis,support_loss)) THEN candidate(periodontitis).

Source: efp; review: pending.

R24 deep_pocket

IF gt(pocket_mm,3) AND kb(buccal_pocket_threshold(3)) THEN deep_pocket.

Source: efp; review: pending.

R25 gingival_inflammation

IF gte(bop_percent,10) AND eq(plaque,yes) THEN gingival_inflammation.

Source: efp_g; review: pending.

Appendix D Acceptance Test Cases

All cases are synthetic. Software results below were reassessed from the source fixtures on 29 September 2026. Clinical expectations remain pending review. TC01-TC14 have diagnostic targets; TC15-TC20 are edge cases. Raw boundary fixtures may deliberately contain child fields hidden by the UI. The routed diagnostic checks independently exclude those fields and preserve each target.

The tables give every explicitly supplied fixture input. Any omitted question is Unknown, never No. Values are the stable Prolog identifiers from the controlled catalogue; checkbox lists identify selected items, while none records explicit absence. Invalid or contradictory fixtures are programmatic boundary tests, not combinations users can freely type into the interface.

D.1 Question identifiers and display labels

Identifier	Question	Stage
age_group	Age group	Setup
dentition	Dentition	Setup
tooth_type	Affected tooth type	Setup
region	Affected region	Setup
tooth_pain	Is tooth pain present?	Step 1
triggers	What triggers the tooth pain?	Step 1
persistence	Pain after the trigger stops	Step 1
spontaneous	Spontaneous tooth pain?	Step 1
sleep_pain	Tooth pain interrupts sleep?	Step 1
biting_pain	Pain when biting?	Step 1
gum_symptoms	Gum symptoms	Step 1
warning_signs	Warning signs	Step 1
jaw_clicking	Jaw clicking?	Step 1
cavity	Cavitated lesion observed?	Step 2
soft_tissue	Softened tooth tissue observed?	Step 2
discoloration	Decay-related surface discoloration?	Step 2
root_maturity	Permanent root maturity	Step 2
thermal_response	Recorded thermal response	Step 2
percussion	Percussion or palpation tenderness?	Step 2
radiographic_caries	Dentist-interpreted coronal carious lesion?	Step 2
apical_pathology	Apical/furcation pathology or pathological resorption?	Step 2
plaque	Plaque or calculus present?	Step 2
prior_destruction	Previous periodontal destruction?	Step 2
pocket_mm	Maximum probing depth	Step 2
cal_mm	Maximum interdental attachment loss	Step 2
buccal_cal_mm	Maximum buccal/oral attachment loss	Step 2
bop_percent	Sites bleeding on probing	Step 2
nonadjacent_teeth	Interdental loss at two nonadjacent teeth?	Step 2
two_teeth	Buccal/oral loss at two teeth?	Step 2
nonperiodontal_causes	Could another local cause account for the attachment loss?	Step 2

D.2 TC01 Caries in a primary tooth

Question identifier	Supplied value
soft_tissue	yes
region	single_tooth
cavity	yes
radiographic_caries	yes
warning_signs	none
percussion	no
apical_pathology	no
gum_symptoms	none
plaque	yes
pocket_mm	3
cal_mm	0
bop_percent	0
tooth_pain	no
spontaneous	no
sleep_pain	no
biting_pain	no
age_group	child
dentition	mixed
tooth_type	primary

Expected response: Dental caries must be included; coexisting candidates permitted.

Actual response: Supported candidate conditions. Candidates: Dental caries.

Missing requests: none.

Message: Candidate conditions only. Clinical knowledge review is pending.

Prohibited outcomes: Unsupported pulpitis or periodontitis.

Result: PASS software. Clinical review: pending.

Source references: [1] and [3]

D.3 TC02 Caries in a permanent tooth in the child group

Question identifier	Supplied value
soft_tissue	yes
region	single_tooth
cavity	yes
radiographic_caries	yes
warning_signs	none
percussion	no
apical_pathology	no
gum_symptoms	none
plaque	yes
pocket_mm	3
cal_mm	0
bop_percent	0
tooth_pain	no
spontaneous	no
sleep_pain	no
biting_pain	no
age_group	child
dentition	permanent
tooth_type	permanent
root_maturity	unknown

Expected response: Dental caries must be included; coexisting candidates permitted.

Actual response: Supported candidate conditions. Candidates: Dental caries.

Missing requests: none.

Message: Candidate conditions only. Clinical knowledge review is pending.

Prohibited outcomes: Unsupported pulpitis or periodontitis.

Result: PASS software. Clinical review: pending.

Source references: [1] and [3]

D.4 TC03 Caries in an adult permanent tooth

Question identifier	Supplied value
soft_tissue	yes
region	single_tooth
cavity	yes
radiographic_caries	yes
warning_signs	none
percussion	no
apical_pathology	no
gum_symptoms	none
plaque	yes
pocket_mm	3
cal_mm	0
bop_percent	0
tooth_pain	no
spontaneous	no
sleep_pain	no
biting_pain	no
age_group	adult
dentition	permanent
tooth_type	permanent
root_maturity	mature
thermal_response	brief

Expected response: Dental caries must be included; coexisting candidates permitted.

Actual response: Supported candidate conditions. Candidates: Dental caries.

Missing requests: none.

Message: Candidate conditions only. Clinical knowledge review is pending.

Prohibited outcomes: Unsupported pulpitis or periodontitis.

Result: PASS software. Clinical review: pending.

Source references: [1] and [3]

D.5 TC04 Reversible pulpitis in a primary tooth

Question identifier	Supplied value
region	single_tooth
cavity	yes
soft_tissue	unknown
radiographic_caries	yes
warning_signs	none
percussion	no
apical_pathology	no
gum_symptoms	none
plaque	yes
pocket_mm	3
cal_mm	0
bop_percent	0
tooth_pain	yes
spontaneous	no
sleep_pain	no
biting_pain	no
persistence	brief
age_group	child
dentition	mixed
tooth_type	primary
triggers	[sweet]

Expected response: Reversible pulpitis must be included; coexisting candidates permitted.

Actual response: Supported candidate conditions. Candidates: Dental caries; Reversible pulpitis.

Missing requests: none.

Message: Candidate conditions only. Clinical knowledge review is pending.

Prohibited outcomes: Unsupported irreversible pulpitis or periodontitis.

Result: PASS software. Clinical review: pending.

Source references: [2] and [3]

D.6 TC05 Reversible pulpitis in an immature permanent tooth

Question identifier	Supplied value
region	single_tooth
cavity	yes
soft_tissue	unknown
radiographic_caries	yes
warning_signs	none
percussion	no
apical_pathology	no
gum_symptoms	none
plaque	yes
pocket_mm	3
cal_mm	0
bop_percent	0
tooth_pain	yes
spontaneous	no
sleep_pain	no
biting_pain	no
persistence	brief
age_group	adolescent
dentition	permanent
tooth_type	permanent
root_maturity	immature
triggers	[cold]

Expected response: Reversible pulpitis must be included; coexisting candidates permitted.

Actual response: Supported candidate conditions. Candidates: Dental caries; Reversible pulpitis.

Missing requests: none.

Message: Candidate conditions only. Clinical knowledge review is pending.

Prohibited outcomes: Unsupported irreversible pulpitis or periodontitis.

Result: PASS software. Clinical review: pending.

Source references: [2] and [3]

D.7 TC06 Reversible pulpitis in a mature permanent tooth

Question identifier	Supplied value
region	single_tooth
cavity	yes
soft_tissue	unknown
radiographic_caries	yes
warning_signs	none
percussion	no
apical_pathology	no
gum_symptoms	none
plaque	yes
pocket_mm	3
cal_mm	0
bop_percent	0
tooth_pain	yes
spontaneous	no
sleep_pain	no
biting_pain	no
persistence	brief
age_group	adult
dentition	permanent
tooth_type	permanent
root_maturity	mature
thermal_response	brief
triggers	[cold,sweet]

Expected response: Reversible pulpitis must be included; coexisting candidates permitted.

Actual response: Supported candidate conditions. Candidates: Dental caries; Reversible pulpitis.

Missing requests: none.

Message: Candidate conditions only. Clinical knowledge review is pending.

Prohibited outcomes: Unsupported irreversible pulpitis or periodontitis.

Result: PASS software. Clinical review: pending.

Source references: [2] and [3]

D.8 TC07 Irreversible pulpitis candidate in a primary tooth

Question identifier	Supplied value
region	single_tooth
cavity	yes
soft_tissue	unknown
radiographic_caries	yes
warning_signs	none
percussion	no
apical_pathology	no
gum_symptoms	none
plaque	yes
pocket_mm	3
cal_mm	0
bop_percent	0
tooth_pain	yes
spontaneous	yes
sleep_pain	yes
biting_pain	no
persistence	lingering
age_group	child
dentition	mixed
tooth_type	primary
triggers	unknown

Expected response: Symptomatic irreversible pulpitis must be included; coexisting candidates permitted.

Actual response: Supported candidate conditions. Candidates: Dental caries; Symptomatic irreversible pulpitis.

Missing requests: none.

Message: Candidate conditions only. Clinical knowledge review is pending.

Prohibited outcomes: Unsupported reversible pulpitis or periodontitis.

Result: PASS software. Clinical review: pending.

Source references: [2] and [3]

D.9 TC08 Irreversible pulpitis in an adolescent permanent tooth

Question identifier	Supplied value
region	single_tooth
cavity	yes
soft_tissue	unknown
radiographic_caries	yes
warning_signs	none
percussion	no
apical_pathology	no
gum_symptoms	none
plaque	yes
pocket_mm	3
cal_mm	0
bop_percent	0
tooth_pain	yes
spontaneous	yes
sleep_pain	yes
biting_pain	no
persistence	lingering
age_group	adolescent
dentition	permanent
tooth_type	permanent
root_maturity	mature
thermal_response	lingering
triggers	[cold]

Expected response: Symptomatic irreversible pulpitis must be included; coexisting candidates permitted.

Actual response: Supported candidate conditions. Candidates: Dental caries; Symptomatic irreversible pulpitis.

Missing requests: none.

Message: Candidate conditions only. Clinical knowledge review is pending.

Prohibited outcomes: Unsupported reversible pulpitis or periodontitis.

Result: PASS software. Clinical review: pending.

Source references: [2] and [3]

D.10 TC09 Irreversible pulpitis in an adult permanent tooth

Question identifier	Supplied value
region	single_tooth
cavity	yes
soft_tissue	unknown
radiographic_caries	yes
warning_signs	none
percussion	no
apical_pathology	no
gum_symptoms	none
plaque	yes
pocket_mm	3
cal_mm	0
bop_percent	0
tooth_pain	yes
spontaneous	yes
sleep_pain	yes
biting_pain	no
persistence	lingering
age_group	adult
dentition	permanent
tooth_type	permanent
root_maturity	mature
thermal_response	lingering
triggers	[cold,hot]

Expected response: Symptomatic irreversible pulpitis must be included; coexisting candidates permitted.

Actual response: Supported candidate conditions. Candidates: Dental caries; Symptomatic irreversible pulpitis.

Missing requests: none.

Message: Candidate conditions only. Clinical knowledge review is pending.

Prohibited outcomes: Unsupported reversible pulpitis or periodontitis.

Result: PASS software. Clinical review: pending.

Source references: [2] and [3]

D.11 TC10 Gingivitis in mixed dentition

Question identifier	Supplied value
tooth_pain	no
spontaneous	no
sleep_pain	no
gum_symptoms	[bleeding,redness]
plaque	yes
warning_signs	none
cal_mm	0
pocket_mm	3
prior_destruction	no
biting_pain	no
cavity	no
percussion	no
age_group	child
dentition	mixed
tooth_type	na
region	general_gums
bop_percent	30

Expected response: Gingivitis must be included; coexisting candidates permitted.

Actual response: Supported candidate conditions. Candidates: Gingivitis.

Missing requests: discoloration, radiographic_caries, soft_tissue.

Message: Candidate conditions only. Clinical knowledge review is pending.

Prohibited outcomes: Unsupported periodontitis or pulpitis.

Result: PASS software. Clinical review: pending.

Source references: [4] and [6]

D.12 TC11 Gingivitis in adolescent permanent dentition

Question identifier	Supplied value
tooth_pain	no
spontaneous	no
sleep_pain	no
gum_symptoms	[bleeding,redness]
plaque	yes
warning_signs	none
cal_mm	0
pocket_mm	3
prior_destruction	no
biting_pain	no
cavity	no
percussion	no
age_group	adolescent
dentition	permanent
tooth_type	na
region	general_gums
bop_percent	40

Expected response: Gingivitis must be included; coexisting candidates permitted.

Actual response: Supported candidate conditions. Candidates: Gingivitis.

Missing requests: discoloration, radiographic_caries, soft_tissue.

Message: Candidate conditions only. Clinical knowledge review is pending.

Prohibited outcomes: Unsupported periodontitis or pulpitis.

Result: PASS software. Clinical review: pending.

Source references: [4] and [6]

D.13 TC12 Gingivitis in the adult group

Question identifier	Supplied value
tooth_pain	no
spontaneous	no
sleep_pain	no
gum_symptoms	[bleeding,redness]
plaque	yes
warning_signs	none
cal_mm	0
pocket_mm	3
prior_destruction	no
biting_pain	no
cavity	no
percussion	no
age_group	adult
dentition	permanent
tooth_type	na
region	anterior_gums
bop_percent	35

Expected response: Gingivitis must be included; coexisting candidates permitted.

Actual response: Supported candidate conditions. Candidates: Gingivitis.

Missing requests: discoloration, radiographic_caries, soft_tissue.

Message: Candidate conditions only. Clinical knowledge review is pending.

Prohibited outcomes: Unsupported periodontitis or pulpitis.

Result: PASS software. Clinical review: pending.

Source references: [4] and [6]

D.14 TC13 Periodontitis in the adult group

Question identifier	Supplied value
dentition	permanent
region	multiple_teeth
tooth_type	permanent
tooth_pain	no
gum_symptoms	[bleeding,redness]
plaque	yes
nonadjacent_teeth	yes
two_teeth	yes
nonperiodontal_causes	no
prior_destruction	unknown
spontaneous	no
sleep_pain	no
biting_pain	no
cavity	no
percussion	no
warning_signs	none
age_group	adult
cal_mm	4
pocket_mm	6
buccal_cal_mm	unknown

Expected response: Periodontitis must be included; coexisting candidates permitted.

Actual response: Supported candidate conditions. Candidates: Periodontitis.

Missing requests: discoloration, radiographic_caries, soft_tissue.

Message: Candidate conditions only. Clinical knowledge review is pending.

Prohibited outcomes: Unsupported intact-periodontium gingivitis or pulpitis.

Result: PASS software. Clinical review: pending.

Source references: [5]

D.15 TC14 Periodontitis in the older-adult group

Question identifier	Supplied value
dentition	permanent
region	multiple_teeth
tooth_type	permanent
tooth_pain	no
gum_symptoms	[bleeding,redness]
plaque	yes
nonadjacent_teeth	yes
two_teeth	yes
nonperiodontal_causes	no
prior_destruction	unknown
spontaneous	no
sleep_pain	no
biting_pain	no
cavity	no
percussion	no
warning_signs	none
age_group	older_adult
cal_mm	6
pocket_mm	7
buccal_cal_mm	unknown

Expected response: Periodontitis must be included; coexisting candidates permitted.

Actual response: Supported candidate conditions. Candidates: Periodontitis.

Missing requests: discoloration, radiographic_caries, soft_tissue.

Message: Candidate conditions only. Clinical knowledge review is pending.

Prohibited outcomes: Unsupported intact-periodontium gingivitis or pulpitis.

Result: PASS software. Clinical review: pending.

Source references: [5]

D.16 TC15 Missing age group

Question identifier	Supplied value
tooth_pain	yes
triggers	[cold]
tooth_type	permanent

Expected response: Additional information needed.

Actual response: Additional information needed. Candidates: none.

Missing requests: age_group.

Message: Complete the required consultation inputs.

Prohibited outcomes: Any supported candidate.

Result: PASS software. Clinical review: pending.

Source references: Implemented boundary and lifecycle requirements

D.17 TC16 Invalid numeric age-group input

Question identifier	Supplied value
age_group	-2

Expected response: Invalid input.

Actual response: Invalid input. Candidates: none.

Missing requests: age_group.

Message: Choose a valid predefined value, including a listed age group.

Prohibited outcomes: Any supported candidate.

Result: PASS software. Clinical review: pending.

Source references: Implemented boundary and lifecycle requirements

D.18 TC17 Missing required tooth information

Question identifier	Supplied value
age_group	child
tooth_pain	yes
triggers	[cold]
tooth_type	unknown
warning_signs	none

Expected response: Additional information needed.

Actual response: Additional information needed. Candidates: none.

Missing requests: tooth_type.

Message: Complete the required consultation inputs.

Prohibited outcomes: Any supported candidate.

Result: PASS software. Clinical review: pending.

Source references: Implemented boundary and lifecycle requirements

D.19 TC18 Contradictory tooth-pain inputs

Question identifier	Supplied value
age_group	adult
dentition	permanent
tooth_type	permanent
tooth_pain	no
spontaneous	yes

Expected response: Conflicting inputs.

Actual response: Conflicting inputs. Candidates: none.

Missing requests: none.

Message: Tooth-pain answers conflict. Clarify whether pain is present.

Prohibited outcomes: Any supported candidate.

Result: PASS software. Clinical review: pending.

Source references: Implemented boundary and lifecycle requirements

D.20 TC19 Jaw-only presentation outside scope

Question identifier	Supplied value
age_group	adult
dentition	permanent
region	jaw_joint
tooth_type	na
warning_signs	none
jaw_clicking	yes
tooth_pain	no
gum_symptoms	none

Expected response: Outside supported scope.

Actual response: Outside supported scope. Candidates: none.

Missing requests: none.

Message: This jaw presentation is outside the supported tooth-pain and gum-symptom scope.

Prohibited outcomes: Any supported candidate.

Result: PASS software. Clinical review: pending.

Source references: Implemented boundary and lifecycle requirements

D.21 TC20 Blank consultation after reset

Procedure: complete TC11 in the UI, choose New consultation, then assess the blank reset state. This fixture represents the resulting empty observation list; Java lifecycle tests separately verify that previous selections and results were cleared.

Question identifier	Supplied value
All questions	Unknown after reset

Expected response: Additional information needed.

Actual response: Additional information needed. Candidates: none.

Missing requests: age_group.

Message: Complete the required consultation inputs.

Prohibited outcomes: Any supported candidate.

Result: PASS software. Clinical review: pending.

Source references: Implemented boundary and lifecycle requirements